

5. A certain computer A with 2.0 GHz CPU clock rate, and CPI of 4.0. The execution time of a program on computer A is 5 seconds.

- a. Compute cycle time.

$$\frac{5 \times 4}{2 \times 10^9}$$

$$= 10 \times 10^{-9} \text{ seconds}$$

SO-A: 2 Points

$$\frac{CPUT}{CCT} = \frac{CCT}{IC \cdot CPI}$$

- b. Compute clock cycles.

$$50 \times 10^9$$

SO-A: 2 Points

- c. Compute instruction count.

$$IC = \frac{\text{Clock Cycles Time}}{CPI}$$

$$= \frac{10 \times 10^{-9}}{4}$$

$$= 2.5 \times 10^9$$

SO-A: 1 Points

- d. Computer B was designed to the program in 2 seconds. Which computer, A or B, is faster and by how much?

SO-B: 2 Points

B faster than A
by $5/2 = 2.5$ times

$$1 \frac{1}{2}$$

SO-B: 1 Point

6. The decimal number 127 is represented as:

- a. $1111\ 1111_8$
- b. $1000\ 0000_8$
- c. EE_H
- d. $0111\ 1111_8$

7. The decimal number equivalent to the following 2's complement number is _____.

$1111\ 1111\ 1111\ 1111\ 1111\ 1111\ 1110\ 0110$.

- a. -6
- b. -25
- c. -26
- d. -2,147,483,622

SO-B: 1 Point

8. The minimum number of bits required to store the hexadecimal number FF is:

- a. 2
- b. 4
- c. 8
- d. 16

SO-B: 1 Point

9. The normalized form of the binary number 0.101_2 is:

- a. 1.01×2^{-1}
- b. 0.101×2^0
- c. $1.01 \times 2^{+1}$
- d. 1.01×10^{-1}

SO-B: 1 Point

10. Let I be the number of instructions in a program, C the number of cycles used by the program, T and clock cycle time (clock period), and R the clock rate. Which of the following does NOT give you the execution time of the program?

- a. $(I \times CPI) / R$
- b. $I \times CPI \times T$
- c. $I \times T$
- d. C / R

SO-B: 1 Point

2. Give the decimal representation of the following single-precision floating point format number:
(Show all calculations.)

110000010111000000000000000000000₂

Sign Exponent Fraction

SO-B: 5 Points

$$\text{Sing} = 1 \text{ (negative)}$$

$$\text{Exponent} = 10000010 = 130 - 127 = 3$$

$$\text{Fraction} = 111$$

So

$$= -1.111 \times 2^3$$

$$= -1111$$

$$= (-15)_{10}$$

$$\boxed{= -15}$$

SO	A		B					C
Question	1	5	1	2	3	4	5	6
Points	5	5	5	5	5	3	2	10
Total	5	0	3	5	5	3	8 8	
Max Marks	5	10	17.5 20					8 10

1. For each of the following multiple choices circle the correct answer:

1. Numbers are stored and transmitted inside a computer in _____.

SO-A: 1 Point

- ☒ a. binary form
- b. code form
- c. decimal form
- d. alphanumeric form

2. Processor's clock speed is measured in _____.

SO-A: 1 Point

- a. bits per second
- b. baud
- c. bytes
- ☒ d. Hertz

3. The size of the exponent field in the IEEE 754 double precision format is _____ bits.

SO-A: 1 Point

- a. 8
- ☒ b. 11
- c. 23
- d. 52

4. Computer performance is improved by:

SO-A: 1 Point

- a. increasing number of clock cycles
- b. increasing response time
- ☒ c. increasing clock rate
- d. increasing CPU time

5. The performance of a computer is defined as

SO-A: 1 Point

- a. execution time
- ☒ b. 1/execution time
- c. clock rate
- d. none of the above

3. Find the binary single precision equivalent to the decimal number: -28.75.
(Show all calculations.)

SO-B: 5 Points

$$-28.75$$

$$= -11100.11$$

$$= -1.110011 \times 2^4$$

$$\text{Sign} = 1$$

$$\text{Exponent} = 127 + 4 = 131 = 10000011$$

$$\text{fraction} = 110011$$

Sign	Exponent	Fraction
1	10000011	110011.....000

$$110000011110110...00$$

4. Find the value of the following operation $a + b$ and represent the result in the scientific notation, where $a = 11.10_{\text{two}}$, and $b = -1.01_{\text{two}}$ (Show all calculations.)

SO-B: 3 Points

$$a = 11.10 = 1.110 \times 2^1$$

$$b = -1.01 = -0.101 \times 2^1$$

$$\begin{array}{r} 1.110 \times 2^1 \\ - 0.101 \times 2^1 \\ \hline \end{array}$$

$$= 1.001 \times 2^1$$

$$a + b = 1.001 \times 2^1$$

1. Embedded are designed to run one application that is normally integrated with HW and delivered as a single system. a. T b. F
2. _____ emphasize delivery of good performance to single users at low cost and usually computing technologies.
a. PCs b. Servers c. Embedded computers d. Parallelism
3. _____ computers are the largest class and span widest range of applications.
a. Desktop b. Server c. Embedded d. Client
4. At one extreme of servers, _____-computers present consist of tens of thousands of processors and many terabytes of memory.
a. PC b. smart c. super d. embedded
5. _____ device is a battery operated with wireless connectivity to the Internet.
a. PC b. PMD c. SaaS d. embedded
6. _____ performs arithmetic operations.
a. Control b. Datapath c. IC d. Input/Output
7. _____ span the widest range in cost and capability.
a. PCs b. Servers c. Embedded d. Processors
8. _____ is a service that is deployed via cloud.
a. Pipelining b. Wafer c. PMD d. SaaS
9. Once found good dies are found, they are connected to the input/output pins of a package, using a process called _____.
a. bonding b. abstraction c. dicing d. yield
10. Make system dependable by including redundant components that can take over when a failure occurs is achieved via _____.
a. redundancy b. parallelism c. pipelining d. prediction